

Qatar-2b

R-1870

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_150320 · created 2026-08-03T10:37:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457101.781086 +/- 7e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.064
Transit depth (Rp/Rs)^2	1.21 +/- 1.4 [%]
Orbital Inclination (inc)	89.28 +/- 1.77
Airmass coefficient 1 (a1)	0.7 +/- 0.47
Airmass coefficient 2 (a2)	0.19 +/- 0.18
Scatter in the residuals of the lightcurve fit is	64.86 %
Best Comparison Star	#2 - [52, 411]
Optimal Aperture	1.63
Optimal Annulus	4.92
Transit Duration (day)	0.0723 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.11 ± 0.064 vs 0.16327 (deviation 0.83σ)
Duration fit vs published	0.0723 d vs 0.0767 d (deviation 1.02σ)
Mid-time O–C	0.3 min
Depth SNR	1.7
Residual scatter	64.86 %

Light curve

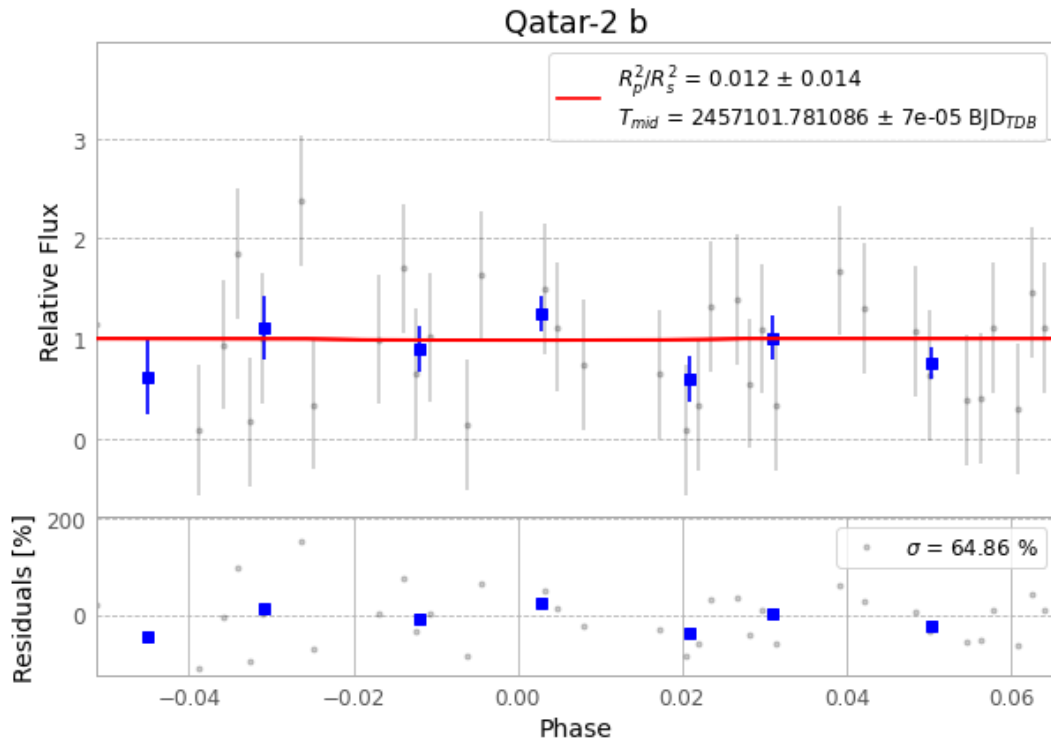


Figure 1 — FinalLightCurve_Qatar-2 b_2015-03-19.png (SHA-256 9dfc2905da500a77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 318], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8888024c579e1953...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar-2150320045412.FITS → Qatar-2150320084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-150320.log (ea396a990a36...), FinalLightCurve_Qatar-2 b_2015-03-19.png (9dfc2905da50...), FinalLightCurve_Qatar-2 b_2015-03-19.csv (053b12bf968b...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:37Z.